SafeStride AI: A Real-Time Intelligent Navigation System with Adaptive Risk Awareness for Visually Impaired

1. Objective

To test the frontend interface for real-time object detection using camera input and verify the accuracy, responsiveness, and usability of predictions.

2. Tools Used

- Web Browser (Chrome)
- Laptop Camera (for real-time input)
- Front-end UI (Camera interface)

3. Front End Details

Parameter		Value
Input source	:	Live Camera feed
Integration	:	`Backend API/predict`
Input	:	Image frames
Output	:	Object label, confidence(object, confidence on UI)

4. Test Cases & Results

Test ID	Image Type	Expected result	Actual Output	Status
TC01	Person	Detect Person	Person Detected(0.53)	✅
TC02	Car	Detect car	Car Detected	✅
TC03	Table and Chair	Detected objects	Table & chair detected	✅
TC04	Wall & floor	No detection	No detection	✅
TC05	Bottle(near camera)	Detect	Detected as hairdryer	❌
TC06	Low light img	Detect	Person detected	✅
TC07	Object at distance	Detect	Detection works but accuracy reduces	❌
TC08	Monkey(from	Detect	Detected as bear	❌

	phone			
TC09	Clear nearby object	Detect correct objects	Objects detected (person,bottle)	✅

5. Observations

- Accurate human detection, even off-camera
- Correct detection for clear objects (e.g., chair)
- Inconsistent results for repeated inputs

6. Bugs Identified

Bug 1:

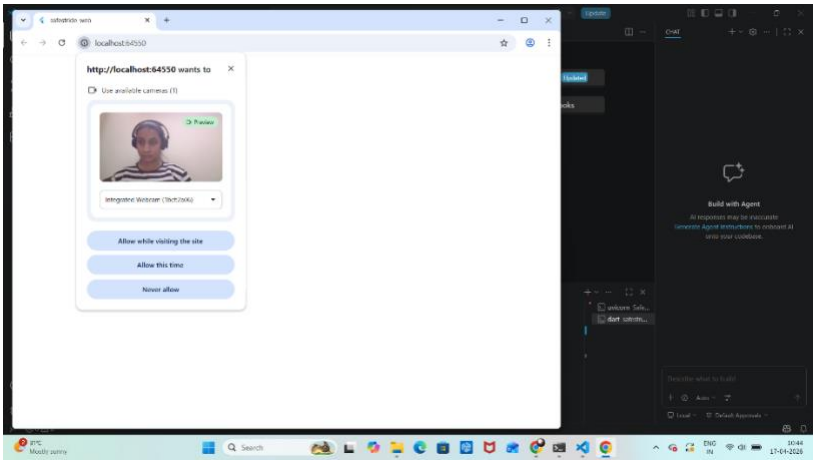
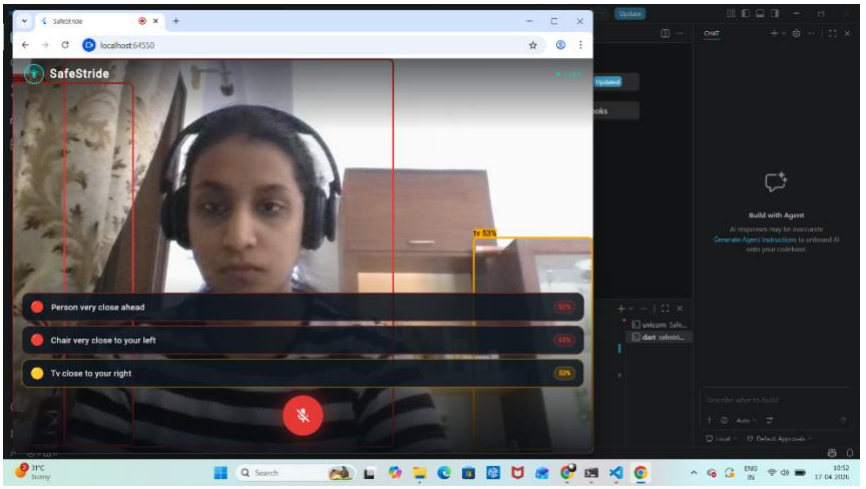
Input: Similar object

Expected: Detect object

Actual: Misclassification of objects

Issue: Reduced accuracy for distant objects

7. Conclusion

The system works well for basic detection but shows issues with accuracy, consistency, and handling edge cases. Improving model training and robustness is needed for better performance.

8. Sample Images